

18.2 A 22nm 1.87ms/Frame Streaming Multi-Speaker ASR Accelerator Leveraging Contextual-Aware Redundancy Skipping with 2D-Writable Microscaling Compute-in-Memory and Similarity-Aware TCAM Design

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Abstract:
This paper presents a DCIM-based accelerator for on-device streaming multi-speaker ASR (MS-ASR), featuring: 1) a context-aware redundancy skipping scheme with online spar... [Show More](#)

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The rising popularity of AI-assisted devices has sparked heightened interest in real-time automatic speech recognition (ASR). Prior single-pass ASR models, such as connectionist temporal classification (CTC) [1]– [2], recurrent neural-network transducer (RNN-T) [3]– [4] and attention-based encoder-decoder (AED) [5]– [6], exhibit high accuracy in terms of word error rate (WER). However, it is challenging to utilize these models for real-time streaming, multispeaker (MS) dialogue, where frequent speaker alternation degrades both inference latency and recognition accuracy [7]. Recent two-pass MS-ASR models combine an intrinsic streaming RNN-T and a high accuracy AED model [8]– [9]. As shown in Fig. 18.2.1, the audio is first encoded by an ASR and a speaker encoder and then attributes to the corresponding speaker by the similarity and clustering module. At the decoding stage, the first RNN-T decoder pass consists of a lightweight joiner and predictor and generates the output sequence simultaneously with the input stream, making it naturally suited for low-latency streaming applications. The second AED pass processes the output list from the RNN-T through an attention mechanism for tokens it generates to achieve higher recognition accuracy.

Previous ASR accelerator chips [10]– [13] focused on single-speaker ASR - none of them consider real-life MS-ASR. MS-ASR has its unique deployment challenges for efficient realtime streaming. First, the sparse context in real conversation induces block-level computation redundancy, which contains 32 – 128 tokens in each block. Silent pauses, represented by blank (Φ) tokens, also create computational redundancy that waste significant energy. Moreover, increasing token-level similarity from 45% to 92% is observed as the transformer layer depth increases (i.e., higher block index) due to the continuity characteristic of speech



signals. Second, two decoding passes introduces distinct requirements for computation and memory. The streaming RNN-T pass decodes speech embeddings with a light-weight joiner (0.23 MB), which has a high invocation frequency consuming 59.2% of memory power [14]. Meanwhile, the attention operator in the transformer-based AED pass requires intense matrix transposes and sparse GEMM, degrading energy efficiency. Third, significant post-processing redundancy arises due to the high overhead of frequent speaker alternation and repetitive evaluations on common words within the N-best rescoring list.

To address the above challenges, this paper presents a digital computing-in-memory (CIM)based accelerator for on-device streaming MS-ASR, featuring: 1) a context-aware redundancy skipping (CARS) scheme with online sparse block prediction, 2) a 2D-writable CIM with memory-intensive static pages and computation-intensive dynamic pages, 3) a similarity-aware ternary content-addressable memory (TCAM) for rapid search of similar speaker embeddings and N-best semantic results. Combining the above techniques, we prototyped the MS-ASR test chip in 22 nm CMOS with a system energy efficiency of 37.50TFLOPS/W in MXFP8 and our chip also shows superior performance on MS-ASR tasks with 1.87 – 7.51ms/ frame and 0.158 – 1.26mJ/ frame compared to prior works [10]– [11].

Figure 18.2.2 shows the chip microarchitecture. It consists of three CIM-based computing clusters, a similarity-aware TCAM system, a RISC-V CPU, a VPU and peripheral circuits. Each computing cluster contains 5 digital CIM macros and a CARS sparse attention unit. The CARS unit contains a similarity-based skipper, a blank-based skipper to dynamically skip repetitive or non-informative (blank) blocks based on online similarity analysis, while a high accuracy decoding trigger is applied to adaptively schedule a rescoring process based on a confidence score of initial transcription. The debug and function controllers generate control signals for the 5 CIM macros to configure their write direction and computation mode. Each CIM macro contains 17 stationary pages (528×256) to store stationary weights and 2 dynamic pages (64×256) to support 2D update and sparse computation. A similarityaware TCAM (SA-TCAM) comprising 32 TCAM banks is used to reduce the latency of speaker attribution and the rescoring process. An active speaker finder (ASF) is added into the TCAM unit to support partial activation of match lines (ML) based on search frequencies and the nearest encoder encodes the address of the closest search results. The reuse-aware N-best scheduler reorders computations based on the N-best score in TCAM, which increases the reuse rate of similar words.

Figure 18.2.3 illustrates the workflow of our context-aware redundancy skipping (CARS). A two-level skipping and one trigger mechanism is employed, which includes (1) similaritybased token block skipping leveraging sparse attention, (2) blank-based token skipping for blank tokens, and (3) context-aware rescoring trigger for non-confidential RNN-T predictions. The input audio frames are first grouped into blocks, and the in-block similarity $\cos(\theta)$ and compressed attention score (CAS) are evaluated. The in-block similarity is approximated in a co-sim threshold block selector based on inequality A1 (displayed in Fig. 18.2.3 right). In the token compression unit, blocks with high similarity are compressed by summing them across the token length. The resulting compressed q and k vectors are then accumulated across the hidden dimensions to compute the CAS. Redundant blocks can be identified and will be marked with “o” in block mask generator if high similarity and low CAS are observed. According to the block index, the blank-based skipper (2)) fetches active blocks and in parallel detects non-blank tokens with a window size of 4. To filter blank frames, a co-sim approximation unit (CAU) is reused to calculate the similarity between the current frame and a reference blank frame, where the left and right part in inequality A2 are generated in adjacent-multiplication and self-multiplication modes of the CAU, respectively. The beam search trigger receives non-blank frames and generates a confidence score for all text candidates. Rescoring is triggered only when there is high ambiguity among the top candidates ($\Delta s < \text{sth}$), otherwise the result is accepted directly. Overall, our CARS scheme reduces MACs by 57-62% with negligible impact on WER.

Figure 18.2.4 outlines the design of our 2D-writable MX-DCIM. The CIM macro supports the microscaling (MX) format [15]– [16] with MX extension modules, which includes an MX block reorder unit, a block max finder for shared scale pre-calculation, and a weight/input aligner for exponent alignment. Since the weights in the joiner module need frequent weightreload, a 17-static-page (16.5KB) DCIM is adopted to store all weights, reducing memory access by 65% and energy consumption by 32%. Meanwhile, the compute-intensive AED pass contains a transformer model with both matrix transpose (QK^T) and sparse GEMM (PV). Although prior work [17] introduced dual adder trees for transpose MVM, this incurs power and area overhead. Our solution is a 2D-writable MX-CIM with two dynamic pages, providing two key benefits to support (1) native in-memory matrix transpose to avoid costly data reordering of intermediate results (K^T) and (2) column-by-column input configuration to send unique input features to non-zero weights in CIM

columns. The two dynamic pages are highly configurable to support four operation modes determined by the SPEN and LW signals. In a standard configuration ($SPEN = 0, LW = 0$), it functions as a conventional CIM with row-wise write (R-write) and input broadcast (input-BC). To enable in-memory transpose, the column-wise write (C-write) mode ($LW=1$) reconfigures the data path, allowing the right DLCs (in [Fig. 18.2.4](#) right) to latch a weight directly from the vertical input feature (IF) ports. Finally, a ‘flash input’ mode ($SPEN=1, LW=1$) is designed for sparsityaware computation, allowing unique input features to be rapidly input into the left DLCs of selected columns. Overall, the 2D-writable MX-DCIM reduces memory access by 72% and saves 43% of energy for the two-pass decoding.

[Figure 18.2.5](#) presents the design of the SA-TCAM. In MS-ASR, speaker embeddings (SimHash compressed) of the same person extracted from different audio clips are often similar but not identical, making the exact matching ineffective. Furthermore, only a few speakers are active in a typical dialogue, making a full search of all stored speaker profiles highly inefficient. The SA-TCAM design supports Hamming distance-based approximate matching and incorporates ASF to dynamically power-gate TCAM entries for inactive speakers. The ASF monitors the search frequency of each speaker by activity down counters, where the entry with minimum distance is selected by the nearest selector (NRS). Once successfully matched, the match signal is asserted by NRS and resets the entry's activity counter to its maximum. Conversely, if the counter expires by reaching zero due to inactivity, the entry's ML is power-gated. This dynamic deactivation of inactive speakers prevents unnecessary searches, yielding a 53% reduction in search power and a 29% reduction in latency. The TCAM is also used to accelerate an N-best rescoring process in full match mode. High-ambiguity candidates from the first pass often exhibit high lexical overlap (i.e., contain many of the same words). Our reuse-aware rescoring (RAR) scheduler leverages this by processing the list from the candidates with highest score to the lowest and storing the decoded key-value pairs of common words in TCAM. When these words reappear (hit) in subsequent candidates, their results are instantly retrieved from the TCAM, bypassing expensive re-computation entirely. The rescoring hit rate is 82% and 71% for single speaker speech and multi-speaker dialogue and this caching mechanism reduces the overall latency by 32%.

[Figure 18.2.6](#) shows the measurement results and comparison table. The chip is fabricated using 22 nm and works at 46.4 – 582MHz with a supply voltage of 0.55 – 1.0V. The MS-ASR tasks are evaluated using two-pass ASR model trained and quantized in microscaling format, in which the RNN-T pass with a size of 0.234 MB is stored in CIM [\[5\]](#), [\[18\]](#). The CARS scheme, STDY-2D-DCIM and SA-TCAM together obtain $12.9\times$ speedup and $13.6\times$ energy savings for clean, single speaker audio and $12.2\times$ speedup and $18.21\times$ energy savings for 3-speaker mixed audio compared to the baseline, with negligible WER increasement. Our accelerator achieves the system efficiency of 37.50TFLOPS/W in MXFP8 format, which is $1.2\times$ better than prior work [\[19\]](#). The per-frame latency and energy for MS-ASR are 7.51 – 1.87ms and 0.158 – 1.26mJ, respectively, as the SoC voltage scales from 0.55 – 1.0V, demonstrating $2.39 - 3.20\times$ speedup and $1.91 - 3.84\times$ energy reduction compared to a prior ASR accelerator [\[11\]](#). [Fig. 18.2.7](#) shows the die photo, voltage-frequency scaling curves and specification table.



Figure 18.2.1:
Overview of two-pass ASR model for multi-speaker and the deployment challenges.



Figure 18.2.2:
Overall architecture of the proposed ASR accelerator and three main features.



Figure 18.2.3:
The two-skip-one-trigger scheme and detailed implementation of CARS.





Figure 18.2.4:
Static and dynamic page-based 2D writable (STDY-2D) CIM macro supporting microscaling format.



Figure 18.2.5:
Speaker-aware approximation search and reuse-aware rescoring with proposed SA-TCAM.



Figure 18.2.6:
Measurement results and performance comparison table.



Figure 18.2.7:
Chip micrograph and specifications.

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